Advanced Programming Practice · 21CSC203P

Unit 5 · Session 1 · SLO 1 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 1 : SLO â€“ 1 | Finite Automata â€” Definition and Types (DFA, NFA)

I. Questions & Answers

1. What is a finite automaton? â€” A theoretical machine with a finite set of states that reads an input string symbol by symbol, transitions between states according to a transition function, and accepts or rejects the string depending on whether it ends in an accepting (final) state.

2. What distinguishes a DFA from an NFA? â€” In a DFA every state has exactly one transition per input symbol (deterministic); in an NFA a state may have multiple transitions for the same symbol, none at all, or Îµ-transitions that move without consuming input.

3. Can every NFA be converted to a DFA? â€” Yes. The subset construction method converts any NFA with n states into an equivalent DFA (with up to 2^n states) that recognizes exactly the same language.